

Linear Transformations and Their Matrices

Linear Algebra - CS 3A and 3B

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BUKC - Spring 2026

What is a Linear Transformation?

Definition

$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if for all vectors $\mathbf{u}, \mathbf{v}$ and scalars c :

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \quad T(c\mathbf{u}) = cT(\mathbf{u})$$

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There exists a unique $m \times n$ matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

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There exists a unique $m \times n$ matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

$$A = (T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots \quad T(\mathbf{e}_n))$$

where $\mathbf{e}_j$ are standard basis vectors.

Geometric Intuition

What linear transformations do:

- Preserve origin: $T(\mathbf{0}) = \mathbf{0}$
- Map lines through origin to lines through origin
- Map parallelograms to parallelograms
- Can **rotate, scale, shear, reflect, project**

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Matrix: $\begin{pmatrix} 2 & 0 \\ 0 & 1/3 \end{pmatrix}$

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Check: $(3, 1)$ rotated 90° CCW gives $(-1, 3)$

General rotation by θ : $A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

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Reflection across x-axis: $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Reflection across y-axis: $A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

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Vertical shear: $T(x, y) = (x, y + 3x)$ matrix $\begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$

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Another: Map to plane $z = 1$: $T(x, y) = (x, y, 1)$? **Not linear!** (fails $T(0) = 0$)

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Check: $x + y = 1 + 2 = 3$, $z = 3$

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Uniform scaling: cI (same factor in all directions)

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Apply to $(2, 0, 5)$:

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Through yz-plane: $(-x, y, z)$ matrix $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

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Apply to $(2, 1, 3)$:

$$A \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \\ 2 + 1 + 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \\ 6 \end{pmatrix}$$

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Example: $(1, 2, 3, 4) \rightarrow (4, 2, 3, 1)$